

The Universal Settlement Layer

The next layer of the internet — focused on data sovereignty and societal flourishing.

This whitepaper presents the Budlum protocol's design for serious evaluation: architecture, consensus model, verifier & trust model, security model and risks. The subject is the protocol; applications, token economics and roadmap live in their own documents.

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1. INTRODUCTION & PROBLEM STATEMENT

Budlum is the **Universal Settlement Layer** — the next layer of the internet — focused on data sovereignty and societal flourishing: it does not replace existing networks; it gathers the certainty they produce into one agreed fact. This section defines the five problems the protocol answers and the limits of current solutions.

1.1 The loss of data sovereignty

Identity, content, reach and revenue moved into the vaults of a few platforms over the past decade. Decentralized storage networks keep bytes available, yet once data reaches a buyer, no on-chain mechanism governs *who may read it, for how long, how many times*. Off-chain promises do not execute; an ownership claim that cannot be cryptographically resolved to its origin is marketing, not verification.

1.2 A fragile societal information space

Who produced a media artifact cannot be verified; impersonation and unattributed reuse are routine. AI outputs entering decision processes cannot be audited. The shared ground of fact that societal flourishing requires — portable identity, owned content, accountable verification — does not exist in today's infrastructure.

1.3 Infrastructure fragmentation

More than 20,000 chains cannot speak to each other. Open networks trade speed for trust; institutional networks trade trust for speed; central-bank digital currency efforts advance in isolated silos. The settlement question between one citizen's digital lira and another's digital euro has no answer today.

1.4 The quantum transition crisis

Every chain in production rests on ECDSA or Ed25519. By NIST's published timeline, cryptographically relevant quantum computers break these signatures between 2030 and 2035. The "harvest now, decrypt later" pattern makes the threat present-tense, and when migration day arrives, incumbent chains must execute it as historically crisis-prone hard forks.

1.5 The verifiability gap

AI inference steers a growing share of economies, yet "who computed what, and did the verifiers agree?" has no on-chain answer. Every application calling an oracle imports a trust assumption it cannot audit.

Positioning — explicit: Budlum's core is a single, **monolithic** settlement ledger; modularity lives at the adapter boundary where domains attach. The protocol is not a competitor to other chains but their settlement counterpart — it aims to make "which consensus?" a meaningless question.

2. ARCHITECTURE OVERVIEW

2.1 Consensus domains

Budlum divides the world into **domains**: the permissionless public domain — the open core network where anyone may verify; the permissioned isolated domain — a fully separate execution environment for institutions, whose membership and rules are institution-defined and whose transactions are invisible to the public network; and external domains that bring their own consensus (PoW, PoS, PoA, BFT, ZK, custom). Only the settlement fact of an isolated domain is anchored — never its transaction contents.

2.2 Inter-domain messaging & the finality adapter

Each domain connects through a **finality adapter**: the adapter produces finality proofs in the domain's own idiom (work proofs, validator-set signatures, light-client checks). Messages travel together with their proofs; a transition that cannot be proven is rejected fail-closed.

2.3 One record: the GlobalBlockHeader

All verified finality proofs fold into a single settlement record, the `GlobalBlockHeader`. Parties on different domains can thereby point to the same fact with the same cryptographic evidence.

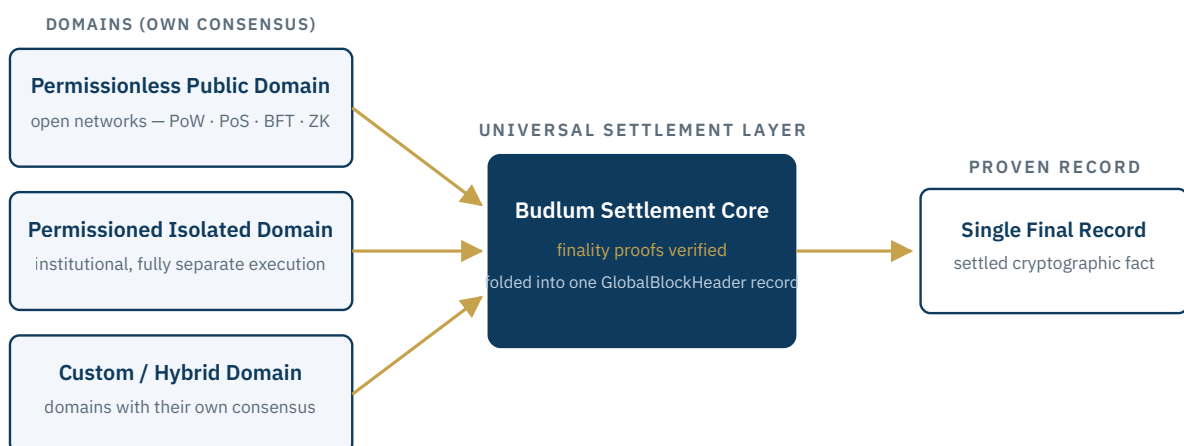


Figure 1 — The Universal Settlement Layer: domains keep their consensus; finality proofs settle into one final record.

3. CONSENSUS MODEL

3.1 Why multi-consensus

No single consensus satisfies every requirement: proof-of-work for cash-like finality, proof-of-stake for performance, proof-of-authority for institutional compliance. Earlier interoperability efforts constrained the answer to one paradigm. Budlum's claim is the inverse: *"Which consensus you use does not matter; I can verify your state."*

3.2 Mandatory hybrid finality

Core finality makes two signature schemes **simultaneously mandatory**: BLS aggregate signatures (today's security) and production-grade Dilithium5 (post-quantum security). The migration is not a crisis-day hard fork; it is already done in the protocol design — systems built on Budlum enter the quantum era without downtime.

3.3 The proof-gated bridge lifecycle

Cross-domain value moves in four steps: **lock** on the source domain, **mint** in the layer, **burn** on the way back, and **unlock** on the target domain. Every transition passes proof gates that reject messages without a finality proof, fail-closed. The \$2.5B history of bridge hacks is answered not with privilege but with a proof requirement.

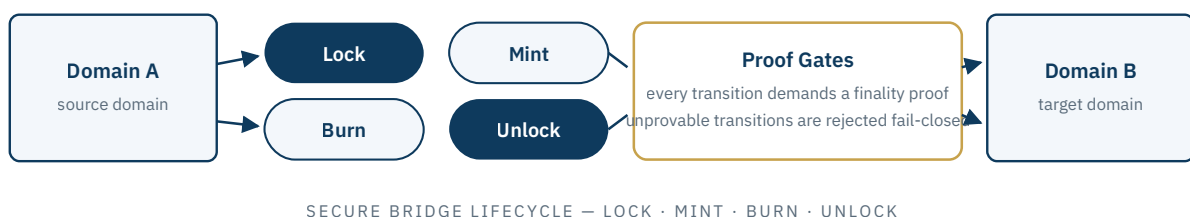


Figure 2 — Bridge lifecycle: every transition demands a finality proof; unprovable transitions are rejected.

4. VERIFIER & TRUST MODEL

4.1 Stake-collateralized registry

Protocol duties are performed through registered roles earned by posting stake collateral in the network's native asset. The model renders any content or transaction accountable: every action has a signer, and every signature has an economic consequence.

4.2 Roles

Role	Responsibility
Validator	Block production and hybrid finality voting
Verifier	Independent verification of registered content and transactions
Relayer	Carrying inter-domain messages together with their proofs
Prover	Generating zero-knowledge proofs
Storage operator	Bonded custody of B.U.D. shards; proof deadlines
AI verifier	Inference attestation; participation in the agreement threshold

4.3 Accountability mechanics

On the storage side, deals are permissionless: anyone posting a positive bond may open custody, anyone may challenge an open deal, and slashing applies only when a proof deadline is missed. On the AI side, incentives are deliberately **soft**: a verifier outside the consensus forfeits the reward share but never loses stake — an accountability line that does not scare new verifiers away.

5. SECURITY MODEL

5.1 Threat model & countermeasures

Threat	Design countermeasure
Bridge attacks	Proof-gated lifecycle; proofless transitions rejected fail-closed
Signing-key compromise	Hardware security module (HSM) signing support on production paths
Quantum break ("harvest now, decrypt later")	BLS + Dilithium5 hybrid finality mandatory in the core path
Unauthorized data access / scope violation	On-chain access objects (scopes, single-use records, revocation) with HPKE key delivery
Corrupt or unknown network payloads	Typed transport; unrecognized payloads rejected fail-closed

5.2 Production-line verification discipline

Verification is treated as the product: no change lands on the main line without passing a mandatory gate line of automated formatting and lint audits, unit and end-to-end test suites, a fuzz round, a non-regressing coverage threshold, full-history secret scanning, supply-chain and dependency audits, and a multi-node network simulation. Constant-time cryptographic comparisons are guarded by dedicated regression tests.

5.3 Independent audits & bug-bounty commitments

Independent external security audits and a public bug-bounty program are committed to before mainnet readiness; findings and remediations will be published openly. This document is commitment-level; program parameters will be announced before launch.

6. RISK FACTORS

6.1 Regulatory risks

The classification of digital assets, data-sovereignty regimes and rules for permissioned institutional domains vary across jurisdictions and may change. The permissioned-domain design offers isolation for compliance; regulatory developments can still shape how the protocol is used.

6.2 Technical risks

Bridge security and the zero-knowledge stack are among the hardest engineering problems in the field; proof gates reduce the risk, they do not eliminate it. Post-quantum parameters continue to mature and may require revision. Software at the v0.3-dev stage is **not audited mainnet software**; it must not be used for real-value production traffic.

6.3 Market risks

Adoption, liquidity and volatility apply to every open network. Nothing in this document implies any return, price target or guarantee; token economics and the roadmap are published in their own documents.

A. APPENDICES

Appendix A Glossary

Term	Meaning
Domain	A self-contained network operating under its own consensus rules, connecting to the layer via finality proofs
Universal Settlement Layer	The Budlum layer that verifies finality regardless of domain architecture and records one agreed fact
Finality	The point at which a transaction is treated as irreversible
GlobalBlockHeader	The single settlement record of all verified finality proofs
Finality adapter	The connector that produces a finality proof in a domain's own idiom
Consensus	A network's method of agreeing on a single state
Stake collateral	Economic collateral locked for roles and accountability
Slashing	Protocol-defined seizure of part of a collateral upon violations
B.U.D.	Broad Universal Database — Budlum's decentralized storage layer
Access object (AccessGrant)	Scoped (single-read / time-boxed / perpetual) on-chain access right; consumption and revocation are recorded
BudZKVM	Budlum's in-tree zero-knowledge virtual machine
BNS / .bud	The Budlum Name Service and its protocol-level portable namespace
Attestation	A verifier's signed approval of an output; a threshold majority finalizes the outcome
HPKE	RFC 9180 hybrid public-key encryption — used for content-key delivery

Appendix B References

1. Budlum core repository (MIT license), github.com/budlum-xyz/budlum — main line at this document's date.
2. *Budlum Paradigm Analysis*, in-house strategic assessment, May 2026.
3. NIST, *Post-Quantum Cryptography Standards* — FIPS 204 (ML-DSA, Dilithium).
4. IETF RFC 9180 — *Hybrid Public Key Encryption (HPKE)*.
5. Boneh, Lynn, Shacham — *Short Signatures from the Weil Pairing* (BLS signatures).

Legal notice

Budlum is MIT-licensed research software with v0.3-dev (controlled devnet candidate) status; it is not audited mainnet software and must not carry real-value production traffic. Nothing in this document constitutes investment advice, an offer or a solicitation. Behaviors described reflect the canonical repository at the document date; topics not yet present (token economics, roadmap) are covered in separate documents.

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